

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	'Low' when temperature goes below minimum temperature and remains in that state until maximum temperature is reached
TP3	5 mV/°C

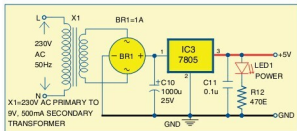


Fig. 3: Circuit diagram of the 5V power supply

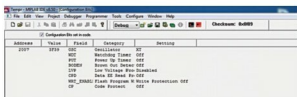


Fig. 4: Configuration bit

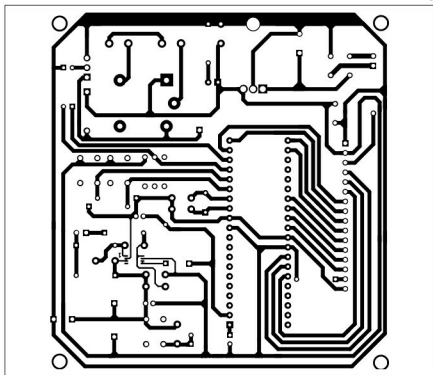


Fig. 5: Combined actual-size, single-side PCB layout for the temperature controller and power supply circuits

There are different types of direct-measurement sensors, for different ranges of temperature (refer Table I).

Thermocouple. Here we have used a K-type thermocouple for direct measurements of up to 1000°C temperature. In K-type thermocouple, the two materials used to form the junction are Chromel (Ni-Cr) and Alumel (Ni-Al). K-type is a low-cost and one of the most popular general-purpose thermocouples. Its operating range is around -250 to +1350°C, with sensitivity of 42µV/°C approximately.

Microcontroller. The heart of the system is microcontroller PIC16F877A, which is a low-power, high-performance, CMOS 8-bit microcontroller. It includes 8Kb Flash memory, 256-byte EEPROM, 368-byte RAM, 33 input/output (I/O) pins, 10-bit 8-channel analogue-to-digital converter (ADC), three timers, watchdog

timer with its own on-chip crystal oscillator for reliable operation, and synchronous PC interface.

Port pins RD0 through RD7 of IC1 are connected to pins D0 through D7 of the LCD. Port pins RB0 through RB2 are connected to register select RS, read/write R/W and enable EN of the LCD. ADC channel RA0 of the microcontroller receives the analogue signal from thermocouple amplifier IC2. Switches S2 through S4 are connected to port pins RC0 through RC2 of IC1. Switches S2 and S3 are used to set the minimum and maximum limits of temperature, respectively. Switch S4 closes to start the ADC function and display the actual temperature. Port pin RC3 controls the heating element. When port pin RC3 goes 'high,' transistor T1 drives into saturation and relay RL1 energises to turn on the heating element.

A 4MHz crystal is connected between pins 13 and 14 of microcontroller IC1 to provide the basic clock frequency. Power-on reset is provided by the combination of resistor R2 and capacitor C1. Switch S1 is used for manual reset. IC2 is a precision instrumentation amplifier with thermocouple cold-junction-compensation circuit. The input signal for IC2 (approximately 42µV/°C) is generated by thermal effects of the thermocouple. IC2 produces output (5mV/°C) directly from a thermocouple signal. With a 5V supply, the 5mV/°C output allows the device to cover nearly 1000 degrees of a thermocouple temperature range. Output of IC2 is connected to ADC input pin RA0 of microcontroller IC1.

The power supply circuit is shown in Fig. 3, where the mains supply is stepped down to 9V, 500mA by transformer X1. This stepped-down AC voltage is rectified by bridge rectifier BR1 and filtered by capacitor C10 before it is fed to IC3. Regulator IC3 provides regulated 5V DC supply. The glowing of LED1 indicates presence of power in the circuit.

Software

The program is written in 'C' language and compiled using Hi-Tech compiler